ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

2025 SEMESTER I FINAL EXAMINATIONS

MULTIMEDIA SYSTEMS (CIT 4541)

INSTRUCTIONS

(1) Time allowed is **Three (3) hours**.
(2) There are Six (6) questions in this paper. **Question ONE (1) and TWO (2) are Compulsory**
(3) Choose **Any other Two (2)** questions from Section B
(4) Each question carries **Twenty (25)** marks.
(5) Calculators are allowed.
(6) The total marks for this Exam is **100**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

SECTION A: COMPULSORY QUESTIONS

Question One

a) A tuning fork vibrates periodically with a period of 0.002 seconds.

i. Calculate the frequency of the sound produced by the tuning fork. **[6 Marks]**

ii. Identify the frequency range (Infrasound, Audible, Ultrasound, or Hyper sound) in which this sound falls. **[4 Marks]**

b) A digital audio system uses 12-bit samples. The signal strength is 4096 and the noise level is 0.5.

i. Calculate the Signal-to-Noise Ratio (SNR) in decibels. **[6 Marks]**

ii. Briefly explain how increasing the number of bits per sample affects quantisation error and SNR. **[4 Marks]**

iii. In a teleconferencing system, a network connection having bandwidth of 100Kbytes/sec is allocated to duplex audio link. What is the maximum sampling rate and resolution in which uncompressed audio data can be transmitted in real-time? **[10 Marks]**

Question Two

a) A music producer is recording a 3-minute podcast using mono audio. The recording is done at a sampling rate of 48,000 samples per second with a resolution of 16 bits. Calculate the total file size of the uncompressed audio recording in megabytes (MB) where 1 MB = 1,048,576 bytes). **[10 Marks]**

b) An audio engineer captures a 2.5-minute stereo sound clip at a sampling rate of 44.1 kHz with a 24-bit resolution. Determine the size of the uncompressed file in megabytes (MB) where 1 MB = 1,048,576 bytes). **[10 Marks]**

c) Record 20 seconds of mono music at 44.1kHz with a resolution, 32 bits, 96kHz. **[5 Marks]**

SECTION B: CHOOSE ANY TWO QUESTIONNS FROM THIS SECTION

Question Three

a) which computer graphics technique can the game development company utilize to simulate realistic lighting and shadows in the game's environments? **[5 Marks]**

b) Calculate the printable image sizes of the following images: a 320 x 240 (pixels) and a 700 x 400 (pixels) images, given that the two images are printed on a 300dpi printer . **[10 Marks]**

c) Discuss the relationship between image, display and output resolutions. **[10 Marks]**

Question Four

Consider the Table below that seeks to compare **three analog television broadcast systems**: **NTSC**, **PAL**, and **SECAM** and answer the questions that follows:

System	Total(scan) Lines	Active Resolution	Vertical Resolution	Horizontal Resolution	Total Pixels
NTSC	525	484	242	330	106,000
PAL	625	575	290	425	165,000
SECAM	625	575	290	465	180,000

a) If PAL has 625 total scan lines and 575 active lines, how many lines are used for non-picture (inactive) purposes such as synchronization? **[10 Marks]**

b) Compare the horizontal resolution of NTSC and SECAM. By how many pixels does SECAM exceed NTSC in horizontal resolution? **[5 Marks]**

c) A photography studio is developing a multimedia system to showcase and sell their photographs online. They want to ensure accurate color representation of their images across different devices and screens. **[10 Marks]**

Question Five

a) i What color representation technique can the photography studio employ to achieve consistent and accurate color reproduction of their photographs on various devices and screens?

b) (2) Distinguish between Chromaticity of PAL and NTSC in colour representation.

Question Three 5

a) The Zambia National Broadcasting Cooperation (ZNBC) is the only broadcaster in Zambia with a wide coverage than any other media house in the country. Because of this, every business organization runs to the cooperation to advertise their products and services. You have been hired by the cooperation to work on adverts production for its clients. Discuss how you would apply your knowledge of multimedia systems to go about creating adverts for ZNBC's clients **[10 Marks]**

b) A marketing agency wants to create an engaging promotional video for a new product. Discuss the multimedia system component would be most suitable for incorporating visual effects and animations into the video. **[5 Marks]**

c) What multimedia system component would be most suitable for incorporating visual effects and animations into the promotional video? Explain your answer. **[10 Marks]**